Housing Discrimination and Algorithmic Bias Analysis

Group Members: Tirth Laheri, Jay Beladiya

Introduction and Historical Context: Racial Covenants are clauses that are embedded into real estate deeds that prohibited the sale and occupancy of homes by any person of color within large parts of Hennepin County, Minnesota between 1910 and 1955. There are over 30,000 racial covenants that have been documented within Hennepin county and also in Ramsey County. These covenants were banned and declared illegal by the Fair Housing Act of 1968.

In today's world the decision to grant a mortgage is done with the help of algorithmic risk assessment systems and machine learning models. These models do not have race as a factor. The model learns from and sets prices according to the property values, neighborhood incomes, and many other factors. We will be analysing two types of bias in this report:

1. Data Bias
2. Algorithmic Bias

Methodology:

Data - The primary dataset combines the 2023 HDMA (Home Mortgage Disclosure Act) lending records for Minnesota with covenant density and demographic/income data. It includes both Hennepin County (22,669 loan records) and Ramsey County (8,782 loan records).

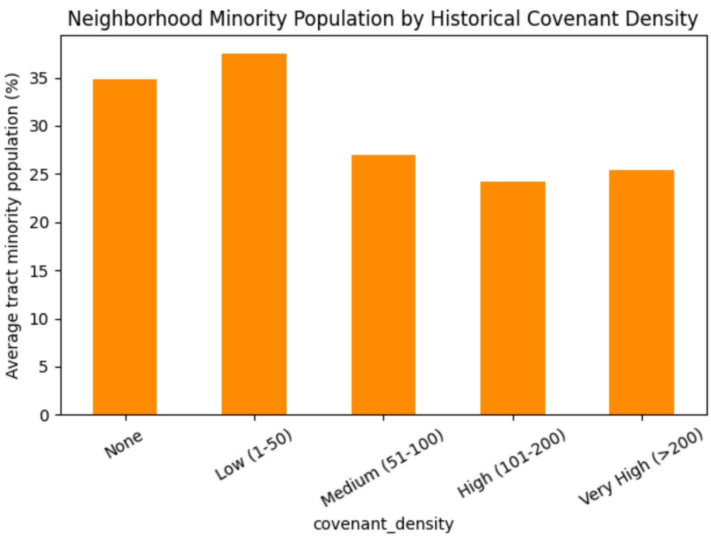
Ramsey County Extension: The data for Ramsey County was already provided in the dataset but the count for 'covenant_count' was zero for every row. So, we combined multiple dataset from the links provided and counted the covenants per tract and sorted them in the same four density bands used for Hennepin county (Low: 1-50, Medium: 51-100, High: 101-200, Very High: >200).

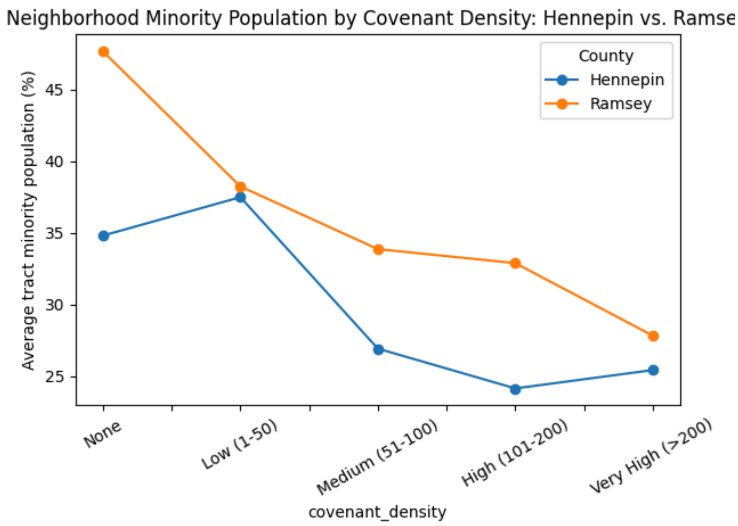
Approach: For data bias, we grouped census tracts by covenant density and compared average property value, minority population percentage and neighborhood income. For algorithmic bias, we compared loan approval rates, interest rates and denial reasons by applicant race and by neighborhood covenant density.

Findings:

1. **Data Bias:** In Hennepin County, property values were noticeably higher in Medium and High covenant density tracts ($567K and $548K average) than in Low-density tracts ($419K), and neighborhood income (relative to the area median) rose fairly steadily with covenant density, from 99% in Low-density tracts to 123% in Very-High-density tracts. Minority population share was highest in tracts with no recorded covenants or only a Low density (35–38%) and notably lower in Medium/High/Very-High density tracts (24–27%).

   In Ramsey County, minority population percentage declined uniformly as covenant density increased — from 47.7% in tracts with no covenants down to 27.8% in Very-High-density tracts — while neighborhood income (relative to area median) rose steadily from 84.7% to 102.5% over the same range.





Looking at the visualizations, in both counties, neighborhoods with a heavier documented history of racial covenants are, on average, whiter and higher-income today than neighborhoods with little or no covenant history. This is the data-bias finding: historical exclusion is still legible in today's neighborhood-level data, roughly 70+ years later.

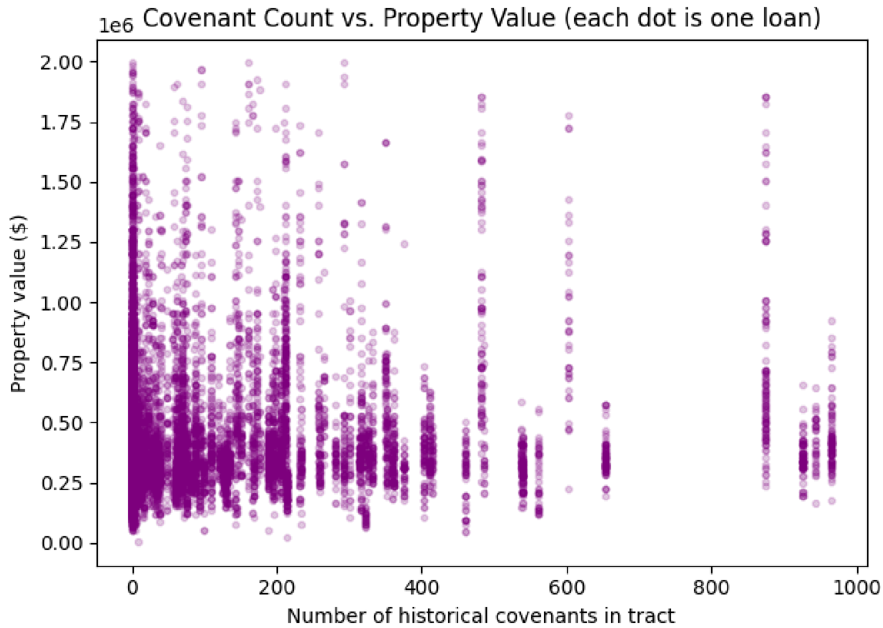


Fig: Covenant Count vs. Property Value

So the data bias pattern is real at the neighborhood category level but it is not a strong, smooth, linear relationship when looking at the raw covenant count.

2. **Algorithmic Bias:** For the purpose of understanding algorithmic bias, the study focused on mortgage lending results based on the merged mortgage and covenant dataset. It looked at differences in loan approval rates, interest rates, and denial reasons between different applicant racial groups and across neighborhoods that have

varying covenant densities. Although lending algorithms are designed to exclude race from the inputs, there is a chance that the algorithms will mirror historical bias patterns found in the data that was used for making decisions.

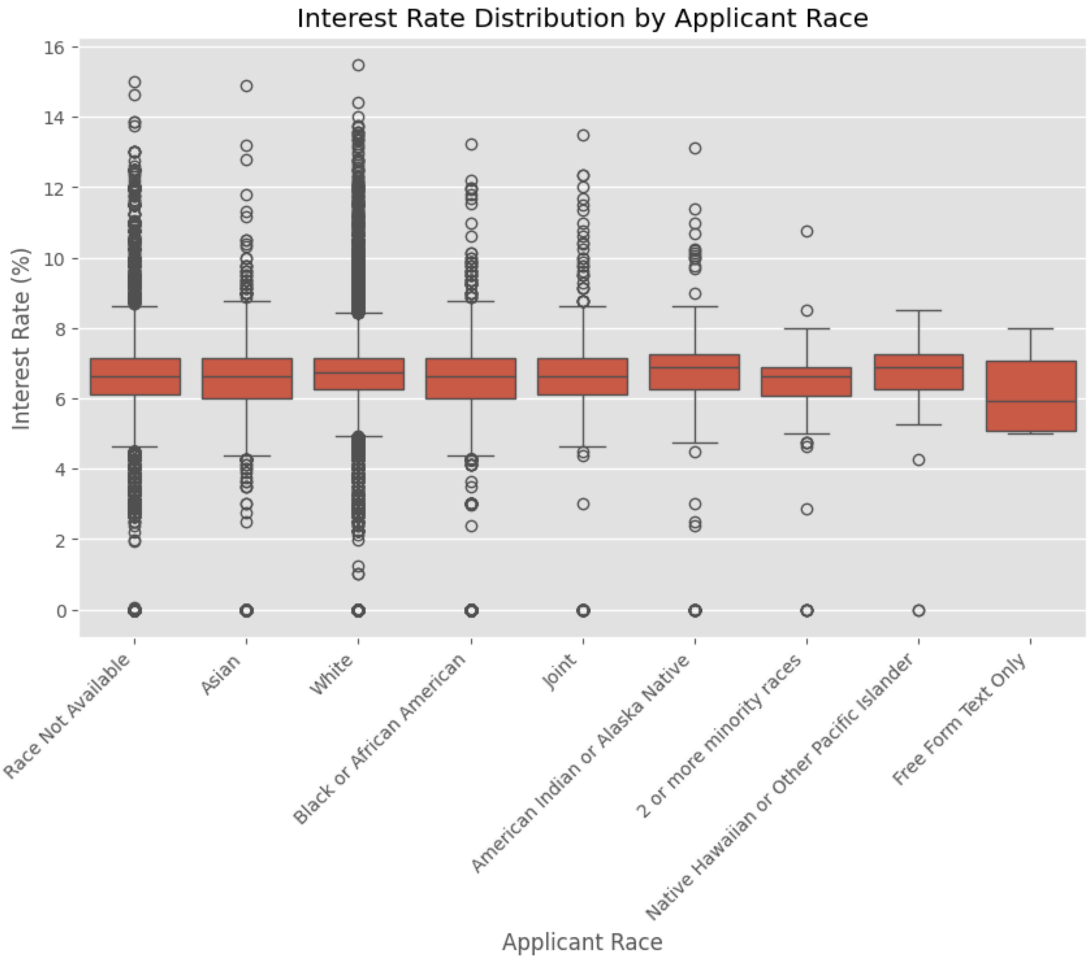


Fig: Interest Rate by Applicant Race

The average of mortgage interest rates for applicant's races were almost the same with medians of nearly 6.4% to 6.9%. Many of the applicants had rates between 5 and 8 percent only. The graphs also showed that the applicants' groups had very high levels of overlap in rate levels. There was some racial group which had very high-interest rates as a minority at more than 10 percent; Yet, the overall differences between them were small so that it is reasonable to say that the racial factor made little difference when it came to the mortgage interest rates of different applicant groups.

Due to Really the interest rates were mainly uniform across different applicant racial groups, we then studied if historical covenant density was a more significant factor in current lending results. What comes next diagrams show comparison of loan approvals and interest rate distributions in neighborhoods that have varying covenant density categories.

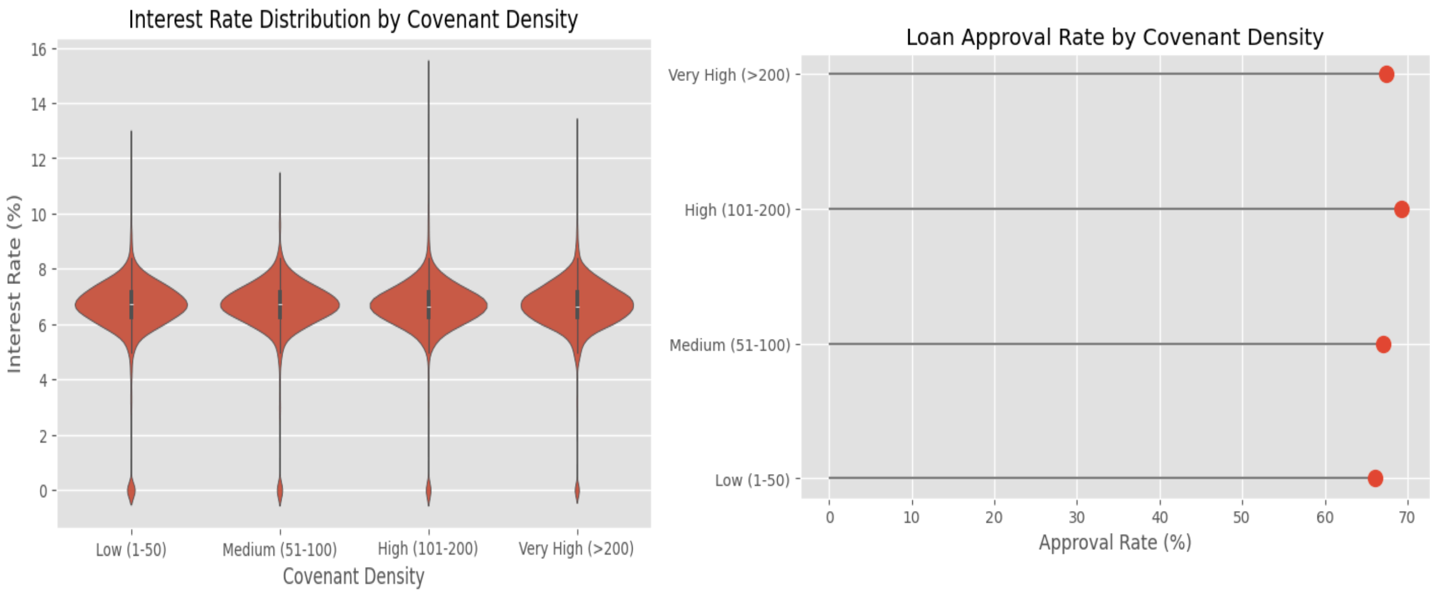
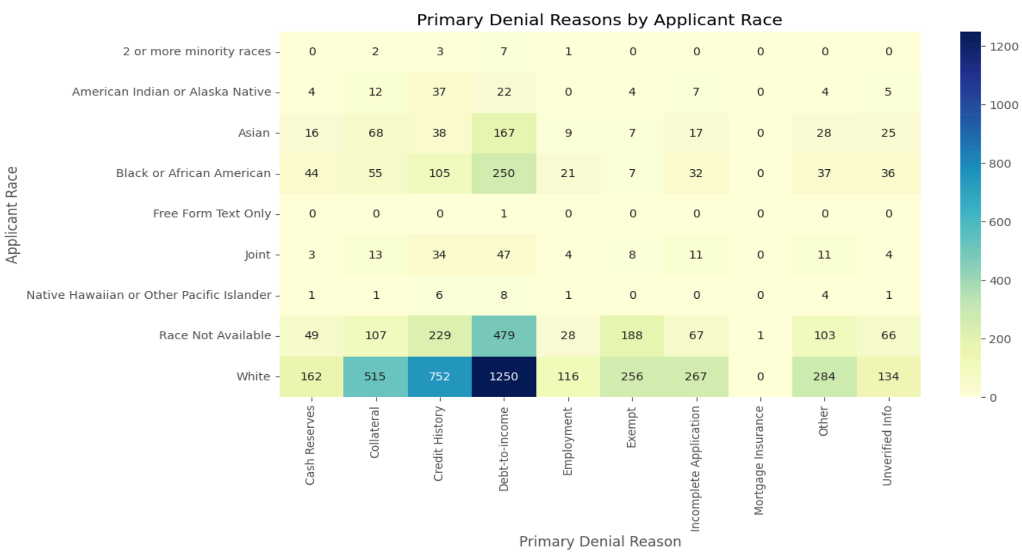


Fig: Interest Rate and Loan Approval Rate by Density

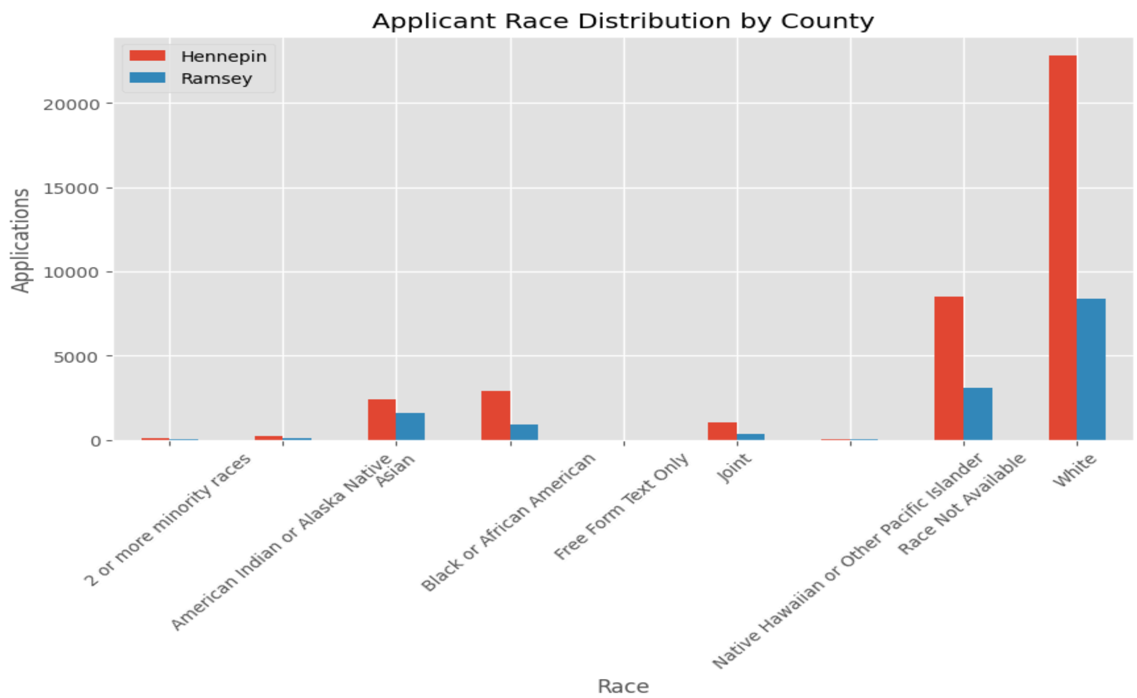
Drawing parallel conclusions from the data, the two images show how home loan results differ between districts with high density contracts (covenants) from their history, unlike other places where such contracts have not been the norm. The graphs indicate that the rate at which loan applications were accepted remained fairly similar in all four zones, from a little over 66% with Low-density neighborhoods to nearly 69% with High-density ones. Also, there was a high level of similarity in the way the interest rate levels were distributed: the median rate was slightly above 6.7% in each case, and most of the loan interest levels fell between 6% and 7.5%, in any covenant situation. Despite the presence of a couple of high-interest rates in each group, there was still a great degree of overlap in the patterns. Basically and overall, the conclusion derived from them is that there's not a very strong link between mortgage loan acceptance rate and interest rate in the 2023 lending dataset on one hand, and the density of covenants that existed before on the other hand.



Debt-to-income ratio was identified as the leading cause of mortgage denial across almost all racial categories, with credit history and collateral being the next most prevalent reasons. Debt-to-income and credit history were the most frequent reasons for loan denial of White applicants, having 1,250 and 752 cases respectively - mainly a reflection of this minority group making the most number of applications. Black or African American applicants' main reason for denial was also debt-to-income with 250 cases, credit history being the second with 105 cases. Across all racial classifications, the dominant denial reason appeared identical, indicating that financially qualifying factors like debt-to-income ratio and payment history play bigger roles in rejecting a loan than the racial background of a borrower.

Overall, this is what the analysis of algorithmic bias revealed: mortgage lending outcomes did not vary much by applicant demographic groups and by neighborhood covenant density. Loan approval rates and interest rates were about the same, the main reasons for denial of the loan throughout the different groups of races were debt-to-income ratio and credit history. That means, it looks like 2023 data is a year in which lending was fairly balanced with only little signs of algorithmic bias. Still, disparities that had been in place for a long time and appear in the form of neighborhood and financial features might have had an indirect effect on the outcome of lending.

3. **Ramsey County Extension:** To extend the analysis beyond Hennepin County, we compared mortgage application patterns between Hennepin and Ramsey Counties using the 2023 Minnesota HMDA dataset. This comparison helps determine whether similar demographic patterns are observed across both counties and provides additional context for interpreting the findings from the main analysis.



While white applicants represented the biggest chunk of mortgage applicants in both Hennepin and Ramsey Counties combined, they are still quite large. For instance 22 700 white applicants were recorded in Hennepin County versus only 8,400 in Ramsey County. Race was not available and Asian applicants were the second biggest category of applicants in both counties. The number of applications recorded for each racial group was higher in Hennepin County which had the bigger pool of applicants overall. Still, there was not much variation in the race-wise distribution of applicants between the two counties.

Discussion Implications for Modern Lending: Despite covenants of the past still shape neighborhood features. Places with a higher covenant density tend to be wealthier and less diverse today. Even though the algorithmic bias analysis revealed only minor differences in loan approval rates and interest rates, it is conceivable historical inequalities will continue to affect lending models mainly through property values, neighboring areas, and credit-related parameters. This shows that taking historical aspects into account is vital in assessing fairness of today's lending systems.

Proposed Mitigation Strategies: Lenders can mitigate bias rooted in historical factors by conducting periodic fairness audits of mortgage approval models that take into account demographic categories and geographic regions. Transparency of automated lending decisions can be improved, model performance is better monitored, and bias assessments carried during the development phase might allow for detecting biased impacts inadvertently. Besides these, financial institutions and governments could keep encouraging the use of equitable lending practices and providing underserved areas with more financial tools.

Constraints and Suggestions for the Future: The findings of this research are based largely on limitations in the HMDA dataset. For example, it lacks information about the full credit history, employment status of the applicants, and debt-to-income ratios of the applicants. It also does not contain any lender-specific underwriting criteria. Also, the historical covenant density was used as a measure of housing discrimination in the past, but it may not have accounted for all elements of the current lending decision. The study suggests further research involving a series of HMDA data years, testing of other counties or states, and use of machine learning fairness tools to explore deeper the ways in which the effects of historical housing discrimination might be embedded in the decision algorithms in automated lending.